MODULE-5
APPLICATIONS OF PHYSICS IN COMPUTING
PHYSICS OF ANIMATION

Animation:

The word Animation has derived from the word Animate, which means to bring life. It is a method of photographing successive drawings, models, or even puppets, to create an illusion of movement in a sequence. The first animated film was made by Emile Chol in 1907; he is referred to as the father of animation.

The Taxonomy of Physics-Based Animation Methods

The physics-based animation and simulation can roughly be subdivided into two groups:

1. Kinematics is the study of motion without consideration of mass or forces.
2. Dynamics is the study of motion taking mass and forces into consideration.

Kinematics and dynamics come in two subgroups:

1. Inverse is the study of motion knowing the starting and ending points.
2. Forward is the study of motion solely given the starting point?

Frames

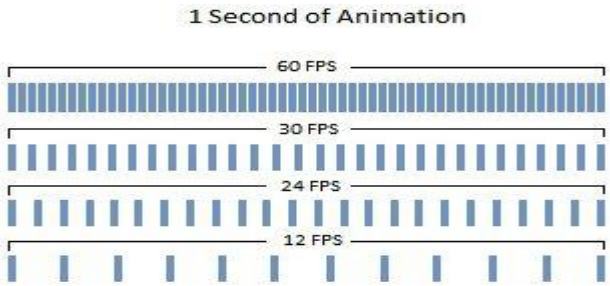
A frame is a single image in a sequence of pictures. A frame contains the image to be displayed at a unique time in the animation. In general, one second of a video is comprised of 24 or 30 frames per second abbreviate as FPS. The frame rate is the speed at which the images are shown in the animation.

The following are the most common types of frames used in animation:

- Keyframe
- Placeholder frames
- Regular frames
- Tweened frames

Frames per Second

There are two standard television images: 30 frames per second (National Television Standards Committee) NTSC for the United States and 25 frames per second (Phase Alternate Line) PAL for Europe. An animated film with 25 frames per second is played on television at 24 frames per second would result in a black bar rolling up the screen. Then Digital Converts are to be used to transfer one speed of the film to another speed of the video.



Size

Size is simply how small or big an element is in relation to other objects within a design. Generally, we use size to make a particular element stand out or to give it importance. However, size becomes a much more powerful design tool when it is considered alongside the scale.

Scaling Properties: Larger or heavier objects move slower while lighter or smaller objects move faster.

When designing characters, you can run into different situationshaving to do with size and scale, such as:

1. Human or animal-based characters that are much larger than we see in our everyday experience. Superheroes, Greek gods and monsters.
2. Human or animal-based characters that are much smaller than we are accustomed to, such as fairies and elves.
3. Characters that need to be noticeably larger, smaller, older, heavier, lighter, or more energetic than other characters.
4. Characters that are child versions of older characters

Proportion and Scale

Creating a larger or smaller character is not just a matter of scaling everything about the character uniformly.

Example: When you scale a cube, its volume changes much more dramatically thanits surface area. Let us say each edge of the cube is 1 unit in length. The area of one side of the cube is 1 square unit, and the volume of the cube is 1 cubed unit. If you double the size of the cube along each dimension, its height increases by 2 times, the surface area increases by 4 times and its volume increases by 8 times.

Weight: Two objects can appear to be different weights by manipulating their timing.

For example, if you were to hit a croquet ball and a balloon with a mallet, the result would be two different actions. The croquet ball would require more force to place it into motion, would go farther, and need more force to stop it. On the other hand, the balloon would require far less force to send it flying, and because of its low mass and weight, it wouldn't travel as far, and would require less force to stop it.

Strength

Strength is the maximum force a muscle or group of muscles can apply against a resistance in a push, pull or lift motion. Body weight is proportional to volume. The abilities of your muscles and bones, however, increase by area because their abilities depend more on the cross-sectional area than volume. To increase a muscle or bone's strength, you need to increase its cross-sectional area.

Motion and Timing animation

Motion is an essential component in games and animations. The motion is governed by newton's laws and kinematic equations. When animating a scene, there are several types of motion to consider. These are the mostcommon types of motion:

1. Linear
2. Parabolic
3. Circular
4. Wave

Timing Animation

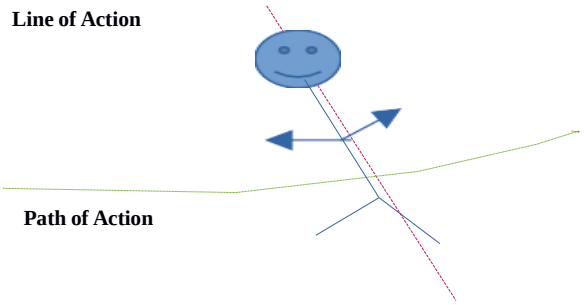
Timing refers to how long an action takes. If the timing is too fast, too slow, too linear, or too long, your animation won't look realistic. Since the film is run at 24 frames per second (FPS), you use this as the building block for your timing. So, if you have an object moving from point A to point B in 24FPS, it takes the object one second to get there.

The functions of timing are to:

1. Create a movement that obeys the laws of physics.
2. Add interest and appeal to your animations.

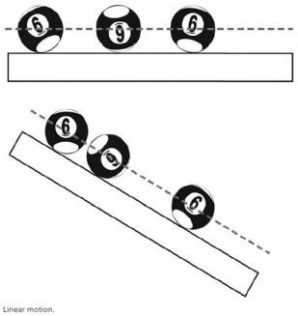
Motion Lines and Paths

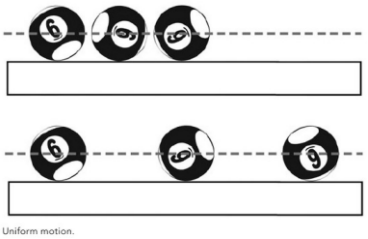
Motionhas a path of action, which indicates the path along which the object or character moves. The path of action refers to theobject's motion in space.



Linear Motion Timing

The linear motion refers to motion in a straight line, always in the same direction. An object moving with linear motion might speed up or slow down as it follows a linear path. A heavy ball rolling on a table or incline is an example of linearmotion.





Uniform Motion Timing

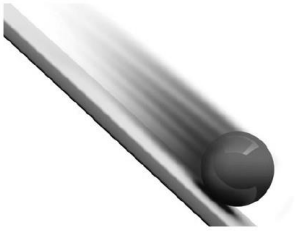
Uniform motion is the easiest to animate because the distance the object travels between frames is always the same. Uniform motion is a type of linear motion with constant speed and no acceleration or deceleration. The object moves the same distance between consecutive frames. Longer the distance between frames, the higher the speed.

Slow in and Slow out

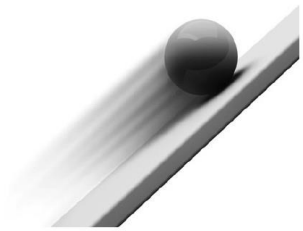
The terms "slow in" and "slow out" are used to describe motion that is accelerating or decelerating. This type of motion is sometimes called ease in or ease out.

1. Slow in/ ease in—The object is slowing down, frequently in anticipation of stopping.
2. Slow out/ ease out—The object is speeding up, often from a static position.

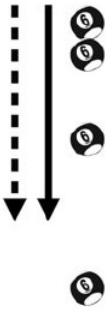
For example, a ball rolling down an incline or dropping straight down is slowing out, as it goes from a still position orslow speed to a fast speed. A ball rolling up an incline is slowing in.



Slowing out: speeding up.



Slowing in: slowing down.



Constant Forces

A constant force is a force that doesn't vary over time.

Examples of constant forces include:

1. Gravity pulling an object to the ground
2. Friction bringing an object to a stop

Constant force and Acceleration

Constant forces result in constant acceleration. Because the acceleration is constant, we can figure out thetiming for such sequences using a few principles of physics.

The resulting acceleration depends on the direction of the force and motion if there is any motion at all, to begin with.

1. When the constant net force is applied to an unmoving object, the result is acceleration.
2. When the constant net force is applied to a moving object in the same direction as the motion, the result is acceleration.
3. When hen constant net force is applied in the direction opposite the existing motion, the result is deceleration (acceleration in the opposite direction).

Forces Exerted by Characters

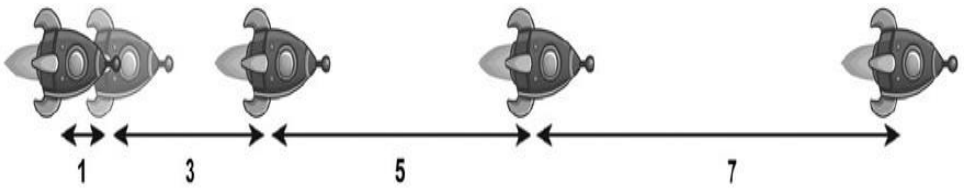
However, one can divide the character's motion into brief time intervals and treat each of these intervals as responding to a constant net force for the purposes of animation. This will make it simpler to determine the timing for each separate segment.

A character walking and pushing a rock is not exerting a constant force throughout the entire sequence, but during each short part of the walk cycle, the net force could be considered to be a different constant value.

The Odd Rule

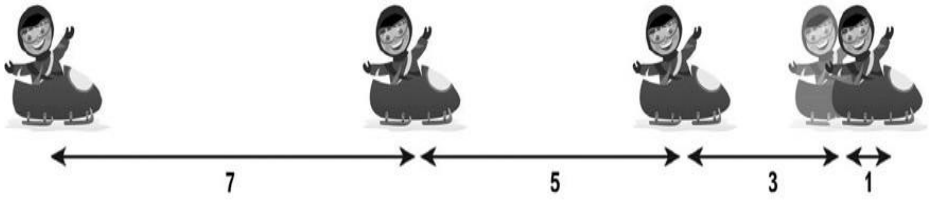
When acceleration is constant, one can use the Odd Rule to time the frames. With this method, one calculates the distance the object moves between frames using a simple pattern of odd numbers. Between consecutive frames, the distance the object moves is a multiple of an odd number. For acceleration, the distance between frames increases by multiples of 1, 3, 5, 7, etc.

The Odd Rule is a multiplying system based on the smallest distance traveled between two frames in the sequence. For a slow-out, this is the distance between the first two frames; for a slow-in, it's the distance between the last two frames. This distance, the base distance, is used in all Odd Rule calculations.



Rocket speeding up using the Odd Rule.

For deceleration, the multiples start at a higher odd number and decrease, for example, 7, 5, 3, 1.



Sled coming to a stop using the Odd Rule.

Odd Rule Multipliers

The Odd Rule in its simplest form, as described above, is just one way to use it. For example, one can instead calculate the distance from the first frame to the current frame and use these distances to place the object on specific frames.

	Multiply by base distance to get distance between:	
Frame #	Consecutive frames	First frame and this frame
1	n/a	0
2	1	1
3	3	4
4	5	9
5	7	16
6	9	25
7	11	36

Odd Rule Scenarios

1. Base Distance Known Speeding up

If the object is speeding up, the first frame distance is the base distance. If one knows the base distance, figuring out the distance the object travels at each frame is pretty straightforward. Just multiply the base distance by 3, 5, 7, etc. to get the distances between consecutive frames, or use squares to multiply the base distance to get the total distance traveled on each frame.

2. Base Distance Known Slowing Down

Suppose one wants an object to slow down, and one knows the distance between the last two frames before it stops. For slow-ins, the base distance is the distance between the last two frames. The solution is to work backward, as if the object were speeding up in the opposite direction. Working Backward, multiply the base distance by 3, 5, 7, etc. to get the distances between each previous frame in the sequence.

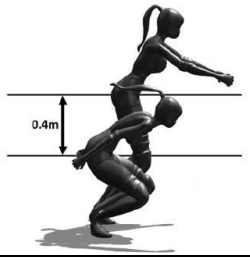
3. Total Distance and Number of Frames Known, Speeding Up

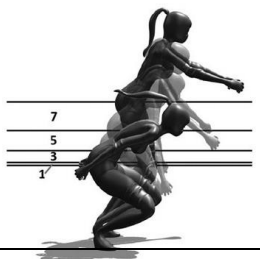
If one wants to know the total distance and the total number of frames, one can find the base

distance with this formula: Base distance = Total distance/(Last frame number – 1) 2

Suppose there is a jump push (takeoff) with constant acceleration over 5 frames, and the total distance traveled is 0.4m. Using the formula above, we find the base distance.

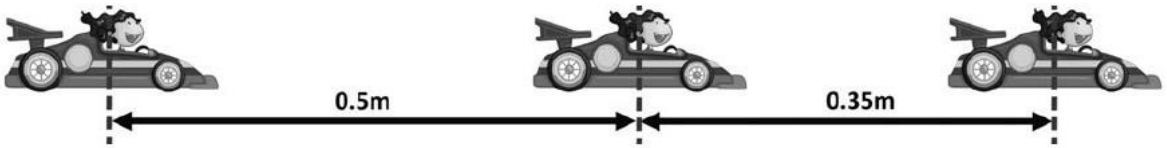
Base distance = 0.4m/(5 – 1) 2 = 0.4m/16 = 0.025m





First Key Distance Known as Slowing Down

Suppose one has a moving object that one wants to slow down, and one has set the first frame of the slow-in to give an idea of the pacing for the sequence.



One feature of the Odd Rule is that the base distance is always half the difference between any two adjacent distances.To find the base distance, one can simply calculate:

(0.5m – 0.35m)/2 = 0.07m

To figure out how many frames are in the slow-in, divide the first distance by the base distance to find out which oddnumber it corresponds to.0.5/0.07 = 7.

This means the first distance corresponds to 7 in the 7, 5, 3, 1 sequence, making the sequence four frames long.

Motion Graphs

A motion graph plots an object's position against time. If one is using animation software, understanding and using motion graphs is a key skill in animating anything beyond the simplest of motions. On a motion graph, the time goes from left to right across the bottom of the graph, while the object's position is plotted vertically against the time. Each axis in3D space (X, Y, Z) has its own line showing the object's position along that axis.

Examples of character animation

Jumping

A jump is an action where the character's entire body is in the air, and both the character's feet leave the ground atroughly the same time. A jump action includes a takeoff, free movement through the air, and a landing.

Parts of Jump

A jump can be divided into several distinct parts:

- **Crouch**: A squatting pose is taken as preparation for jumping.
- **Takeoff:** The character pushes up fast and straightens their legs with their feet still on the ground. The distance from the character's center of gravity (CG) in the crouch to the CG when the character's feet are just about to leave the ground is called the push height. The amount of time (or the number of frames) needed for the push is called the push time.
- **In the air**: The amount of time the character is in the air from takeoff to the apex is called the jump time. If the takeoff pose and the landing pose are similar, then the jump height and jump time are about the same going up as they are going down.
- **Landing**: The distance from the character's CG when her feet hit the ground to the point where the character stops crouching is called the stop height. The stop height is not always exactly the same as the push height.

Calculating Jump Actions

When working out the timing for a jump, one will need to first decide on:

1. Jump height or jump time
2. Push height
3. Stop height
4. Horizontal distance the character will travel during Jump

Example:

Jump height = 1.2m

Jump time for 1.2m = 0. 5 seconds

Jump time at 30fps = 0. 5 * 30 = 15 frames

Jump Magnification

The JM is the ratio of the jump height to the push height.

$$JM = \frac{Jump\ Height}{Push\ Height}$$

Since you already know the jump height and push height, you can calculate the JM. Then you can use the JM tocalculate other aspects of the jump.

Example:

Jump Height = 1m

Push Height =

0.33m

JM = Jump Height/Push Height = 3

Jump Magnification and Acceleration

Jump Magnification is in fact an exact ratio that tells one how much the character has to accelerate against gravity to getinto the air. The JM, besides being the ratio of jump-to-push vertical height and time, is also the ratio of push-to-jump vertical acceleration. Opposite the other ratios: while a longer jump time means a shorter push time, a higher jump acceleration means a much, much higher push acceleration. Knowing about this can help you make more informed decisions about your push timing.

To see how this works, let's look at the formula for JM and relate it to acceleration:Jump Time Jump Height

$$JM = \frac{Jump\ Time}{Push Time} = \frac{Jump\ Height}{Push\ Height} = \frac{Push\ Acceleration}{Jump\ Acceleration}$$

The magnitude of jump acceleration is always equal to gravitational acceleration, with deceleration as the characterrises and acceleration as it falls.

$$JH = \frac{Push\ Acceleration}{Jump\ Acceleration} = \frac{Push\ Acceleration}{Gravitational\ Acceleration}$$

Push Time

The JM also gives you the ratio of the jump time to the

push time.JM = Jump Time/Push Time

Working a little algebra, we can express the equation in a way that directly calculates

the push time:Push Time = Jump Time/JM

Example:

JM = 3

Jump Time: 15 frames

Push Time = 15/3 = 5 frames

Walking

Walks feature all the basics of mechanics while including personality. The ability to animate walk cycles is one of the most important skills a character animator needs to master.

Strides and Steps

A step is one step with one foot. A stride is two steps, one with each foot. Stride length is the distance the character travels in a stride, measured from the same part of the foot. Step and stride length indicate lengthwise spacing for the feet during a walk.

Gait is the timing of the motion for each foot, including how long each foot is on the ground or in the air. During a walk, the number of feet the character has on the ground changes from one foot (single support) to two feet (double support) and then back to one foot. You can plot the time each foot is on the ground to see the single and double support times over time. A normal walking gait ranges from 1/ 3 to 2/3 of a second per step, with 1/2 second being average.

